IVF Bovine Blastocyst Classification and Selection

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Abstract

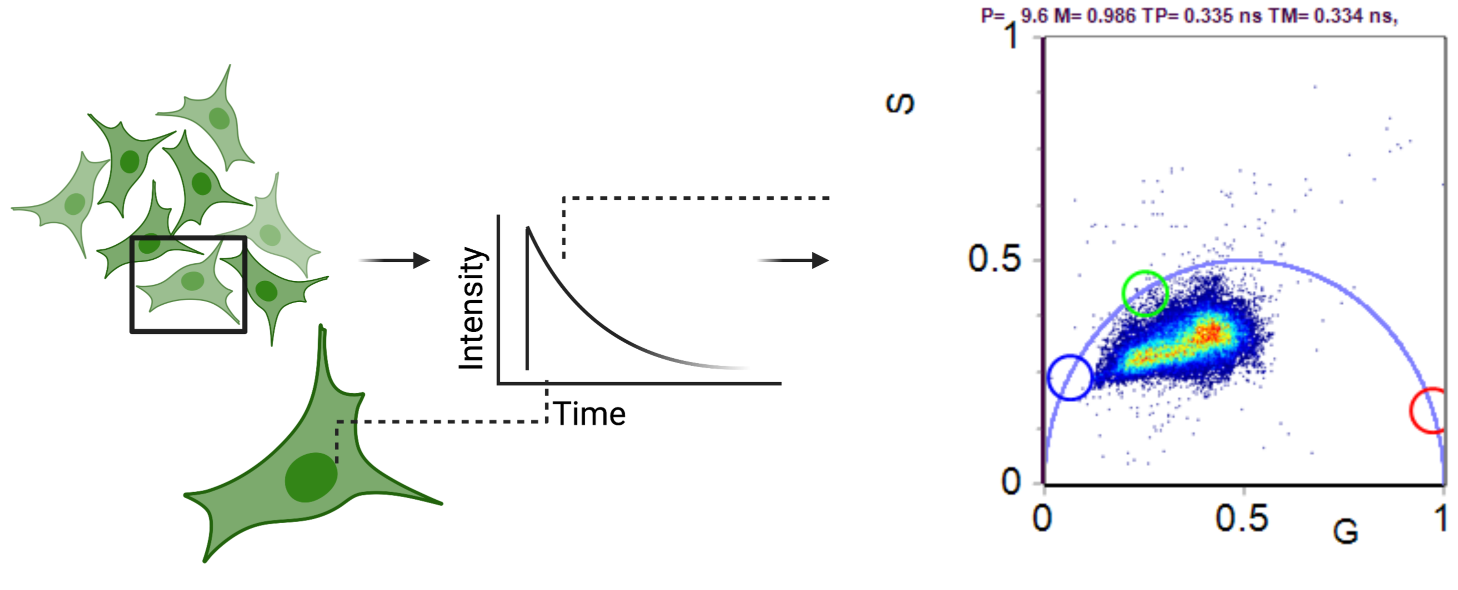
The current technology for in vitro fertilization (IVF) has a long history of merely 40 years. The addition of all live births from either fresh or frozen embryos obtained from a single oocyte collection is known as the IVF productivity rate.[1] The ability to select the best quality embryo is the key to eliminate the number of oocytes generated, procedural cycles for potential failed attempts, and lowering cost and maximize efficiencies. Here we proposed a solution where we focus on developing advanced non-invasive imaging methods to map the energy metabolism of embryos using fluorescent lifetime imaging (FLIM).[3] Combining multiphoton excitation, hyperspectral imaging, and machine learning approach to minimize instrumenting costs, minimize laser exposure to the bovine embryos, and maximize predictably results for embryo selection whether fresh or frozen. Multi-parametric profiling of the entire embryo shows its morphological and metabolic features is then generated, showing significant differences for in vitro produced embryos pf different embryo quality grades.

NADH FLIM

Phasor Transformation

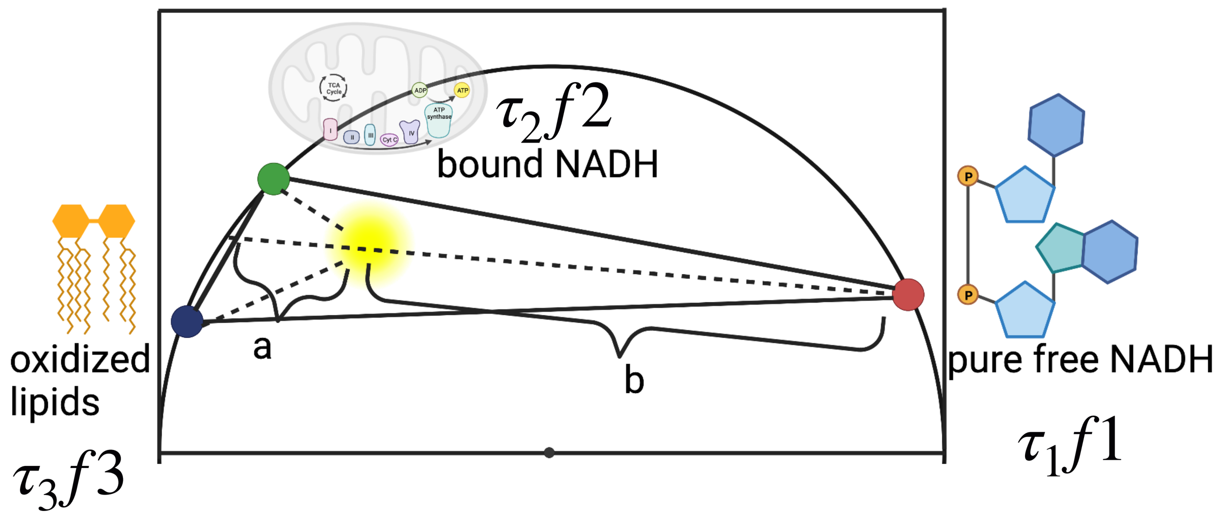
$$S_n(\omega) = \frac{\int_0^T I(t)sin(nt\omega)dt}{\int_0^T I(t)dt} \qquad g_n(\omega) = \frac{\int_0^T I(t)cos(nt\omega)dt}{\int_0^T I(t)dt}$$

Where $\omega = 2\pi f$ is the angular laser repetition frequency
In the case of free/bound NADH, endogenous fluorophores emission coming from mitochondria will be analyzed.



Three Component Analysis

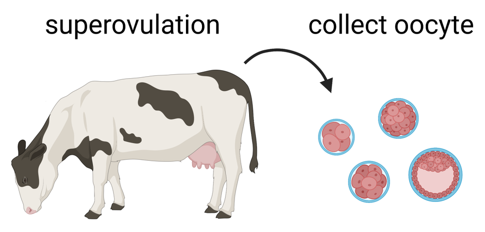
The contribution of the total phasor is made of three components: Free/bound NADH and oxidized lipids. The contribution of each components is the highlighted for samples that are considered healthy or unhealthy.



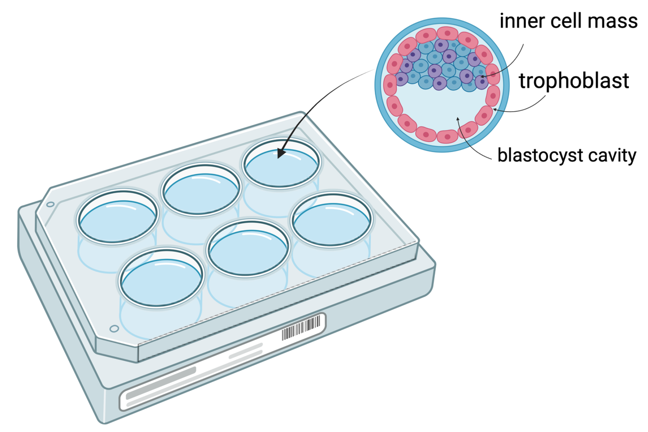
$f1, f2, f3$ are the fractional contribution from each component. It could be calculated using the distance difference in frequency domain.

$$f = \frac{b}{(a+b)}$$

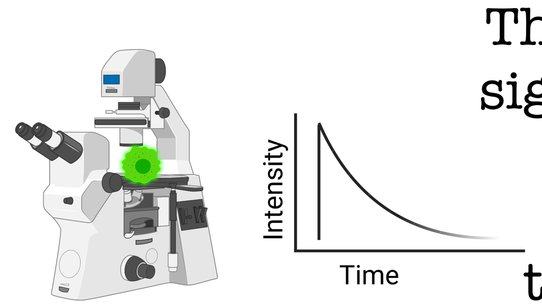
Materials



Oocytes from super ovulated cows are collect for imaging. A range of morula till late blastocyst are examined.

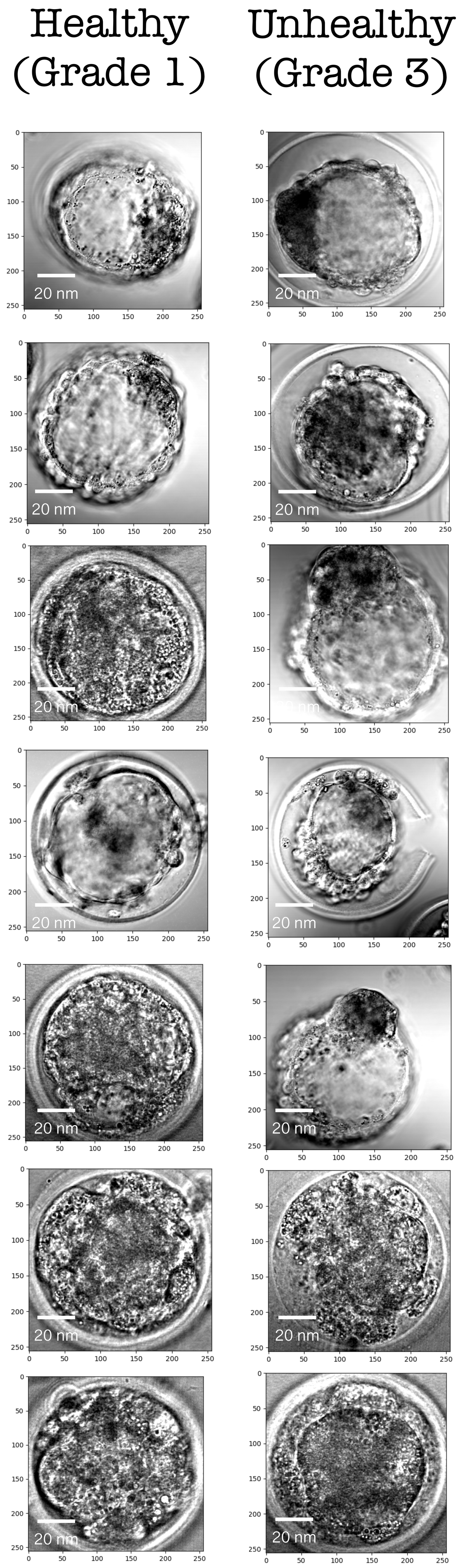


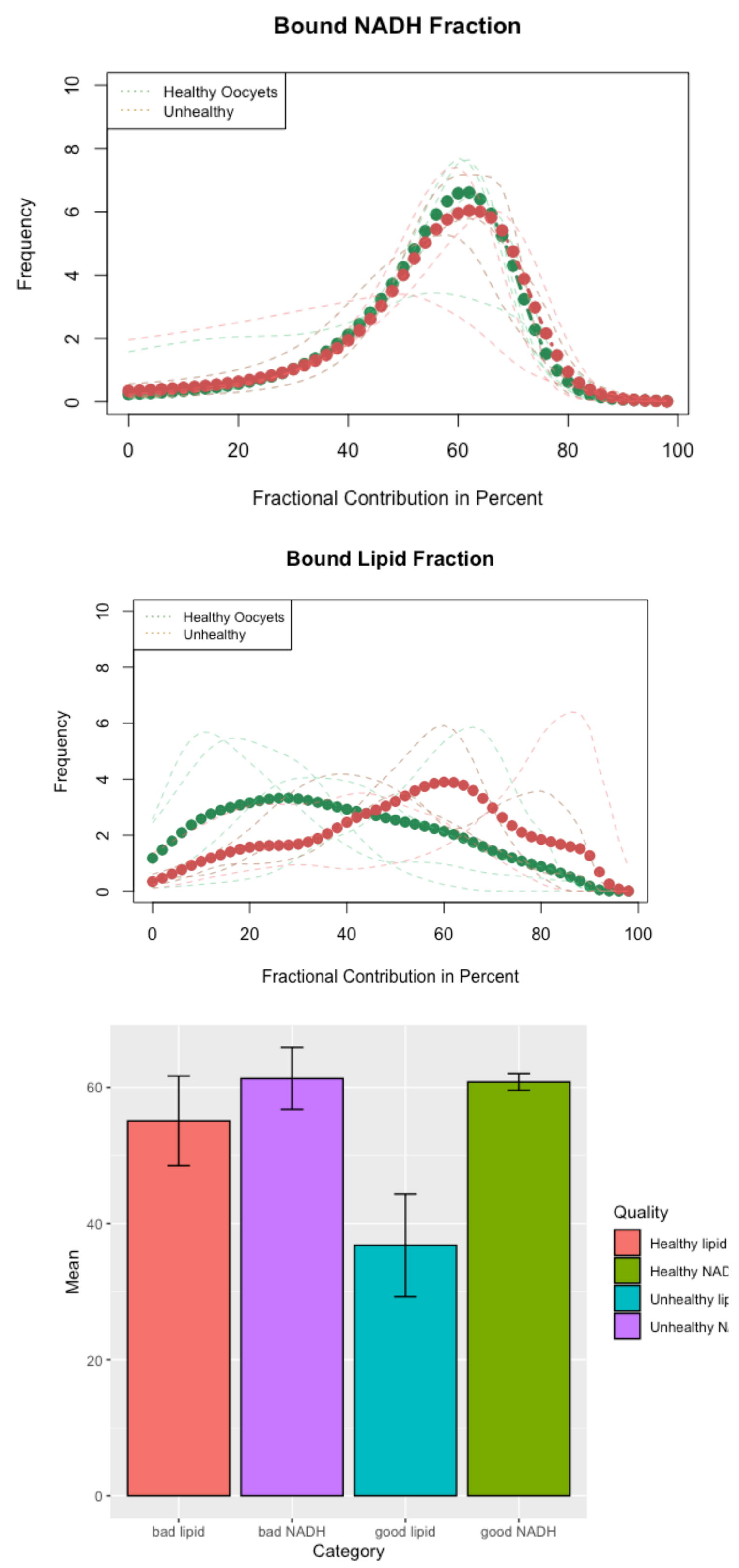
Blastocyst are plated on to imaging dish in group of 3 for imaging

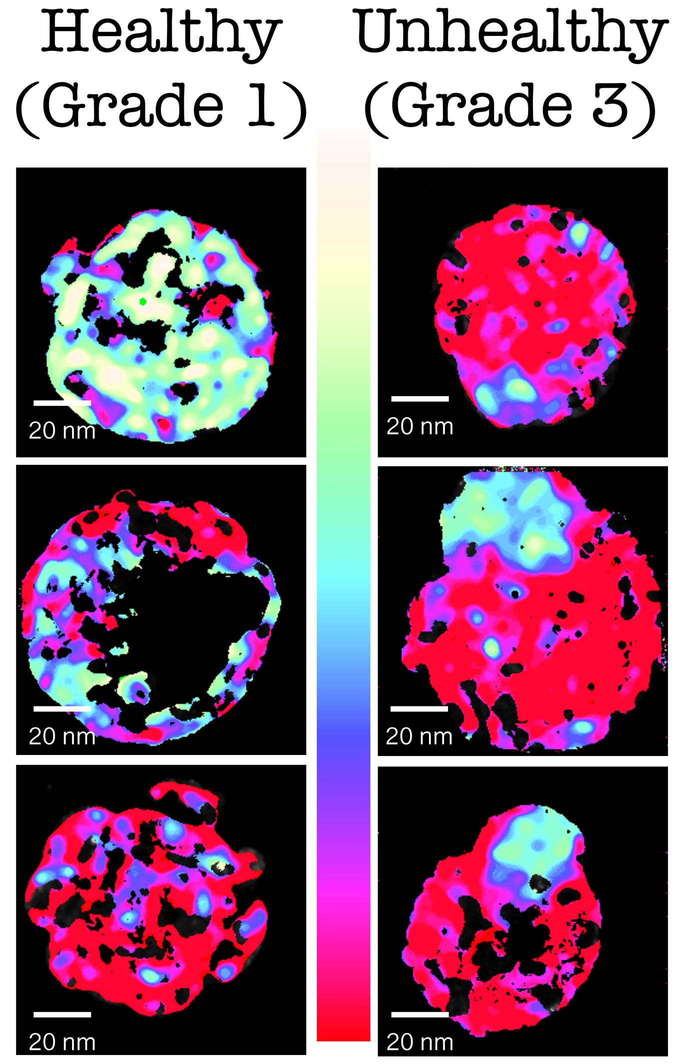


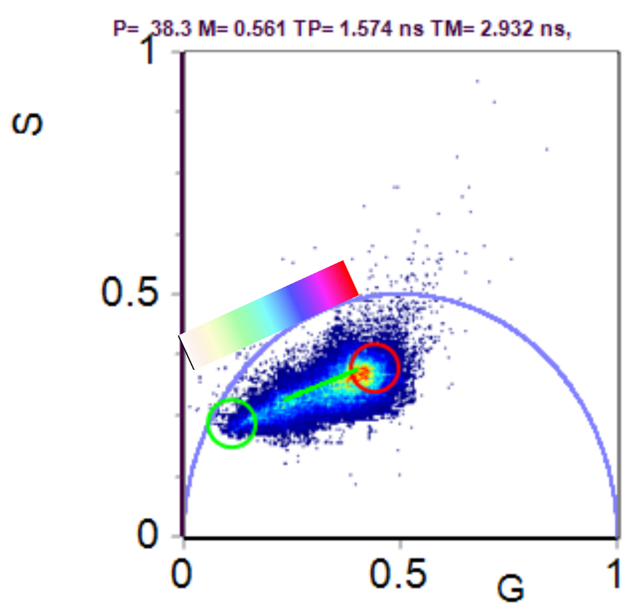
The endogenous fluorescent signals are the collected and analyzed. The overall phasor location shifts toward oxidized species .

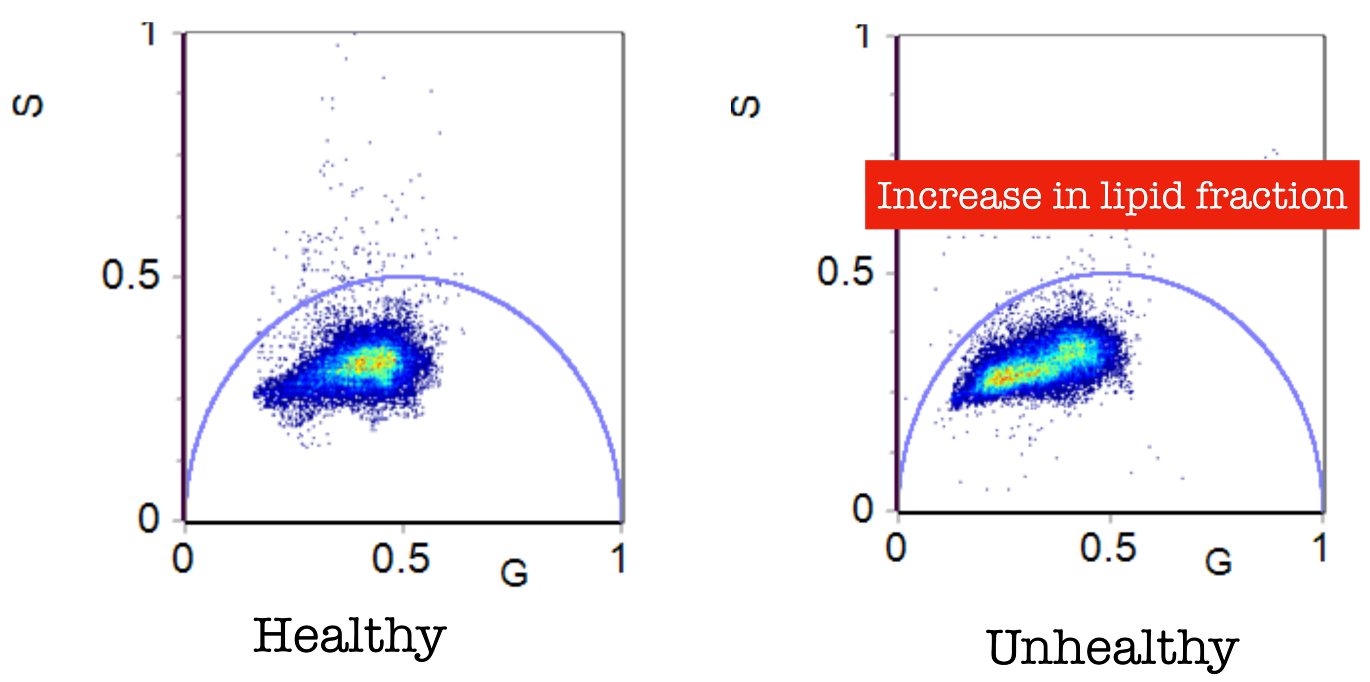
Result











Conclusion & Acknowledgment

- Healthy (grade1) blastocyst has significant less amount of oxidized lipids compare to unhealthy group(grade3)
- Inner cell mass have a longer lifetime compare to trophectoderm

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